

PROJECT SPECIFICATION

Nimbrin Thermal Annex - Thermal and Electrical Work

Project: Nimbrin Thermal Annex | Owner: Orvessa Civic Energy Office

SPEC GUARD DEMO FIXTURE - ALL CONTENT IS FICTIONAL

This invented training specification contains deliberate navigation noise.

No real project, owner, manufacturer, vendor, or product data appears here.

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DOCUMENT NAVIGATION NOTES

Technical requirements are concentrated in Sections 26 05 19 and 26 24 16.

Sections 01, 02, and 28 contain administrative or discipline-specific decoys.

Cross-references identify the governing section and paragraph without replacing it.

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SECTION 01 11 00 - SUMMARY OF WORK

PART 1 - GENERAL

1.1 SUMMARY

- Provide administrative direction for the fictional thermal annex work.
- Coordinate equipment data with Section 26 24 16, Paragraph 1.2.
- This section establishes no electrical rating or enclosure requirement.

1.2 WORK INCLUDED

- Schedule the work so the owner can occupy the existing learning rooms.
- Protect adjacent finishes, stored materials, and temporary access routes.
- Use the project record procedures in Section 01 33 00, Paragraph 2.1.

1.3 COORDINATION

- Resolve conflicts between the fictional drawings and field conditions before ordering.
- Record coordination questions in the submittal register.

SECTION 01 11 00 - SUMMARY OF WORK

PART 3 - EXECUTION

3.1 PROJECT SEQUENCE

- Complete selective demolition before opening the new equipment route.
- Keep temporary power available until the panelboard work is accepted.
- Coordinate shutdowns with the fictional owner representative.

3.2 CLOSEOUT

- Submit record photographs and a final list of unresolved coordination items.
- Include the product package only where another section requires it.

3.3 ADMINISTRATIVE NOTE

- The paragraphs on this page are boilerplate and contain no cut-sheet performance requirement.

SECTION 01 33 00 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUBMITTAL ADMINISTRATION

- Identify the project, specification section, equipment tag, and revision on each submittal.
- Keep the submitted page order intact when a package contains multiple products.
- Do not use a marketing page as the governing technical schedule.

1.2 ACTION SUBMITTALS

- Submit product data, dimensions, connection information, and factory declarations.
- Cross-reference each proposed value to its source page in the package.

1.3 INFORMATIONAL SUBMITTALS

- Provide installation instructions and a final field verification record.
- This administrative list does not change the technical requirements in Division 26.

SECTION 01 33 00 - SUBMITTAL PROCEDURES

PART 2 - PRODUCTS

2.1 SUBMITTAL PACKAGE

- Assemble one readable package with an index, product identifiers, and revision dates.
- Separate alternate products from the product selected for the equipment schedule.
- See Section 26 24 16, Paragraph 2.2 for the selected panelboard requirements.

2.2 REVIEW MARKINGS

- Mark the proposed product, accessories, and options without using approval language.
- State a variance in plain language when a submitted value differs from the specification.

2.3 FILE FORMAT

- Provide text-searchable pages in the order shown by the package index.
- No requirement on this page is a cut-sheet performance value.

SECTION 01 78 23 - OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 DESCRIPTION

Collect operation narratives for installed systems and retain them in the project record.
Use the equipment identifier from Section 26 05 53, Paragraph 2.1.
This section is boilerplate and does not prescribe cut-sheet ratings.

1.2 QUALITY ASSURANCE

Review the record set for legibility, indexing, and file naming consistency.
Do not infer a product requirement from an operation narrative.

SECTION 01 78 23 - OPERATION AND MAINTENANCE DATA

PART 2 - PRODUCTS

2.1 RECORD MEDIA

Provide a searchable electronic record and one printed index for the owner.
Use durable labels on record binders and storage media.

2.2 CONTENT INDEX

List the product package, test records, maintenance intervals, and training notes.
Reference Section 01 33 00, Paragraph 2.1 for the package index format.

2.3 EXCLUSION

This boilerplate section contains no requirement relevant to a cut-sheet rating, listing, material, or dimension.

SECTION 02 41 19 - SELECTIVE DEMOLITION

PART 1 - GENERAL

1.1 SUMMARY

Remove abandoned partitions and redundant cable supports only where shown in the fictional drawings.
Protect existing thermal equipment that remains in service.

1.2 SUBMITTALS

Submit a demolition sequence and a list of temporary protection measures.
No equipment product value is established by this section.

1.3 CONDITIONS

Verify concealed conditions before cutting, drilling, or removing supports.
Notify the owner before an unexpected condition changes the sequence.

SECTION 02 41 19 - SELECTIVE DEMOLITION

PART 3 - EXECUTION

3.1 PREPARATION

- Isolate the work area and maintain clear access to occupied rooms.
- Use dust control and collect removed material for proper disposition.

3.2 REMOVAL

- Remove only the items identified by the fictional demolition schedule.
- Patch openings and leave the remaining support surfaces ready for new work.

3.3 BOILERPLATE NOTE

- This section is a navigation decoy and contains no requirements for the submitted panelboard package.

SECTION 26 05 00 - COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 SUMMARY

Coordinate electrical work with the thermal annex room layout and access routes.
Provide supports, labels, and protection for equipment furnished under Division 26.
See Section 26 24 16, Paragraph 3.1 for panelboard installation.

1.2 DEFINITIONS

The terms selected assembly, equipment package, and field record refer to this fictional project only.
A reference to a package page does not replace a specified requirement.

1.3 SUBMITTALS

Submit coordination drawings and a list of proposed accessories before fabrication.

SECTION 26 05 00 - COMMON WORK RESULTS FOR ELECTRICAL

PART 2 - PRODUCTS

2.1 GENERAL MATERIALS

Use new, undamaged materials compatible with the equipment installation environment.
Coordinate finish colors and field labels with Section 26 05 53, Paragraph 1.2.

2.2 REFERENCE RATING SCHEDULE

The following schedule is a coordination aid; governing values appear in the named section.

REFERENCE RATING SCHEDULE

ITEM	NOMINAL VALUE	COORDINATION NOTE
Service frequency	60 Hz	confirm with source equipment
Control circuit	24 VDC	see Section 26 05 19
Equipment depth	280 mm	coordinate with room layout
Label height	18 mm	use for field identification
Record copies	2 sets	one owner set and one field set

SECTION 26 05 19 - LOW-VOLTAGE CONDUCTORS

PART 1 - GENERAL

1.1 SUMMARY

Provide conductors, terminals, preparation, torque records, and circuit labels for the panelboard.
Coordinate termination information with Section 26 24 16, Paragraph 2.1.

1.2 SUBMITTALS

Submit conductor data, terminal data, and the field torque record before landing conductors.
State material, insulation temperature, and terminal range for each conductor group.

1.3 DELIVERY AND STORAGE

Keep conductor ends dry and protected from crushing, abrasion, and unauthorized cutting.

SECTION 26 05 19 - LOW-VOLTAGE CONDUCTORS

PART 2 - PRODUCTS

LAYOUT NOTE - TWO-COLUMN SPECIFICATION LAYOUT

2.1 TERMINALS

Terminal insulation shall be rated 80 deg C minimum for continuous service.
Use listed terminals that accept the installed conductor range.
Provide plated contact surfaces where the terminal schedule requires them.

2.2 CONTROL ACCESSORIES

Provide a 24 VDC control circuit for accessory indications.
Locate auxiliary terminals behind the removable barrier.

2.3 MATERIAL HANDLING

Protect conductor ends from moisture and impact before landing.

2.4 IDENTIFICATION

Mark each conductor at both ends with the circuit designation.
Use the identification rules in Section 26 05 53, Paragraph 2.1.

2.5 FACTORY DATA

Submit the terminal schedule with the equipment data package.
See Section 26 24 16, Paragraph 2.2 for panelboard coordination.

2.6 QUALITY RECORDS

Record the final torque value after each termination is complete.

SECTION 26 05 19 - LOW-VOLTAGE CONDUCTORS

PART 3 - EXECUTION

3.1 INSTALLATION

Prepare and land conductors without nicking insulation or deforming the conductor.
Install the field termination schedule after the selected assembly is identified.

3.2 FIELD RECORDS

Record the actual conductor group, terminal identifier, and final torque value.
See Section 01 33 00, Paragraph 2.3 for searchable record formatting.

CONDUCTOR TERMINATION SCHEDULE

CIRCUIT GROUP	CONDUCTOR	TERMINAL RECORD	FIELD NOTE
Service	copper, 95 mm2	T-01 through T-04	record final torque
Feeder A	copper, 50 mm2	T-05 through T-08	identify phase
Feeder B	copper, 35 mm2	T-09 through T-12	protect spare ends
Control	copper, 2.5 mm2	T-13 through T-16	use 24 VDC label
Spare	copper, 6 mm2	T-17 through T-20	cap unused ends

SECTION 26 24 16 - PANELBOARDS

PART 1 - GENERAL

1.1 SUMMARY

Provide the selected panelboard assembly for the thermal annex distribution room.
Include enclosure, bus, protective devices, branch positions, and factory declarations.
Coordinate conductor landing with Section 26 05 19, Paragraph 2.1.

1.2 RELATED REQUIREMENTS

Section 26 05 00 governs common supports and coordination information.
Section 26 05 53 governs equipment identification and nameplates.

1.3 QUALITY ASSURANCE

Provide factory data that identifies the selected model and its installed options.

SECTION 26 24 16 - PANELBOARDS

PART 1 - GENERAL

1.4 ACTION SUBMITTALS

Submit the complete product package, including ratings, dimensions, construction, and declarations.
Identify the selected product separately from any alternate product pages.

1.5 INFORMATIONAL SUBMITTALS

Submit installation instructions, factory test records, and a spare-position schedule.
Cross-reference each schedule entry to the product page that carries the value.

1.6 DELIVERY

Deliver the assembly with removable barriers protected and the equipment tag visible.

SECTION 26 24 16 - PANELBOARDS

PART 2 - PRODUCTS

2.1 SERVICE DUTY

- Provide a fault-duty rating of 50 kA symmetrical at 480 V.
- Coordinate the incoming conductor arrangement with Section 26 05 19, Paragraph 3.1.

2.2 ENCLOSURE

- The enclosure shall carry environmental classification E-4.
- The cabinet finish shall be graphite gray.

2.3 BUS CONSTRUCTION

- Main bus shall be copper with a continuous tin-finished surface.
- Provide a full-size neutral and a dedicated equipment grounding bar.

SECTION 26 24 16 - PANELBOARDS

PART 2 - PRODUCTS

2.4 LISTING

Provide an assembly listed under the QL-4 panelboard listing.
Submit the declaration page with the selected product schedule.

2.5 ACCESS AND SPACE

Maintain 1000 mm clear working space in front of the selected panelboard.
Keep the barrier removal path clear of fixed building elements.

2.6 BRANCH CAPACITY

Provide three spare 20 A branch positions for future circuits.
Identify spare positions separately from installed active circuits.

PANELBOARD ACCESSORY SCHEDULE

ACCESSORY	REQUIRED PROVISION	COORDINATION
Main disconnect	front accessible	see Paragraph 3.1
Neutral bar	full size	bonding at source
Ground bar	dedicated	bonding jumper record
Branch schedule	printed	submit with package

SECTION 26 24 16 - PANELBOARDS

PART 3 - EXECUTION

3.1 INSTALLATION

Set the panelboard plumb, anchor it to the prepared support, and preserve service access.
Install barriers and identify all active and spare branch positions.

3.2 FIELD QUALITY CONTROL

Inspect the enclosure, bus, labels, and branch schedule before energization.
Record factory and field test results in the closeout package.

3.3 PROTECTION

Protect the completed assembly from dust and impact until the room is accepted.

SECTION 26 27 26 - WIRING DEVICES

PART 1 - GENERAL

1.1 SUMMARY

Provide wiring devices for the adjacent teaching rooms where shown by the fictional drawings.
This section does not govern the panelboard assembly.

1.2 SUBMITTALS

Submit device schedules, color samples, and room-by-room installation notes.

1.3 QUALITY

Use new devices from one coordinated family and protect them from construction damage.

SECTION 26 27 26 - WIRING DEVICES

PART 3 - EXECUTION

3.1 INSTALLATION

Install devices square, level, and flush with the finished wall surface.
Label each device plate by room and circuit identifier.

3.2 FIELD TESTING

Test each device for operation and record the result in the room schedule.

3.3 NAVIGATION DECOY

This section contains no requirement relevant to the Vantrel panelboard cut-sheet package.

SECTION 28 05 00 - COMMON WORK RESULTS FOR ELECTRONIC SAFETY

PART 1 - GENERAL

1.1 SUMMARY

Coordinate pathways and power for electronic safety equipment shown in the security drawings.
This section is independent of the Section 26 24 16 panelboard product data.

1.2 SUBMITTALS

Submit device addressing, pathway diagrams, and software configuration notes.

1.3 DEFINITIONS

Use the fictional project tag format shown in Section 26 05 53, Paragraph 2.1.

SECTION 28 05 00 - COMMON WORK RESULTS FOR ELECTRONIC SAFETY

PART 3 - EXECUTION

3.1 PATHWAY COORDINATION

Keep security pathways separated from the thermal control pathways where shown.
Mark penetrations before concealment and record the final route.

3.2 TEST RECORDS

Submit point-to-point test records and a list of open commissioning items.

3.3 EXCLUSION

No cut-sheet rating, listing, material, clearance, or quantity is established here.

SECTION 28 31 00 - FIRE DETECTION AND ALARM

PART 1 - GENERAL

1.1 SUMMARY

Provide detection and alarm devices for the thermal annex as shown in the fictional life-safety drawings.
This section does not govern distribution panelboard construction.

1.2 RELATED WORK

Coordinate power provisions with Section 26 27 26 and pathway work with Section 28 05 00.

1.3 SUBMITTALS

Submit device lists, sequence narratives, and testing forms.

SECTION 28 31 00 - FIRE DETECTION AND ALARM

PART 3 - EXECUTION

3.1 INSTALLATION

Install detection devices at the locations shown and protect them during finishing work.
Use the room names from the fictional floor plan.

3.2 TESTING

Test alarm notification, supervision, and power-loss reporting before turnover.

3.3 BOILERPLATE NOTE

These requirements are discipline-specific decoys for the navigation evaluation.

SECTION 26 05 33 - RACEWAYS AND BOXES

PART 1 - GENERAL

1.1 SUMMARY

Provide raceways and boxes for conductors serving the thermal annex equipment.
Coordinate entry points with the selected panelboard dimensions.

1.2 SUBMITTALS

Submit raceway layouts, support details, and a list of penetrations.

1.3 QUALITY

Use fittings suited to the installed raceway and protect open ends from debris.

SECTION 26 05 33 - RACEWAYS AND BOXES

PART 3 - EXECUTION

3.1 ROUTING

Route raceways to preserve the panelboard service clearance stated in Section 26 24 16, Paragraph 2.5.
Provide accessible pull points and avoid conflict with removable barriers.

3.2 SUPPORT

Anchor raceways to the prepared structure and leave room for inspection.

3.3 CLOSEOUT

Record final raceway routes on the project closeout drawings.

SECTION 01 79 00 - DEMONSTRATION AND TRAINING

PART 1 - GENERAL

1.1 SUMMARY

Provide orientation for the owner representative on normal operation and routine inspection.
Training topics do not modify the product requirements in Section 26 24 16.

1.2 SUBMITTALS

Submit a training agenda, attendee list, and sign-in record.

1.3 SCHEDULING

Schedule the demonstration after the field test record is complete.

SECTION 01 79 00 - DEMONSTRATION AND TRAINING

PART 3 - EXECUTION

3.1 DEMONSTRATION

Demonstrate normal operation, safe access, and response to a simulated branch interruption.
Do not energize a circuit while a barrier or cover is removed.

3.2 RECORDS

Provide a short demonstration record with the date and attendees.

3.3 EXCLUSION

This administrative section is a decoy and contains no cut-sheet comparison requirement.

SECTION 26 05 53 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

Provide identification for the panelboard, branch circuits, conductors, and field records.
Coordinate labels with the equipment tag shown in Section 26 24 16, Paragraph 1.1.

1.2 SUBMITTALS

Submit a label schedule and a final marked-up equipment directory.

1.3 MATERIALS

Use legible, durable labels that remain attached after routine cleaning.

SECTION 26 05 53 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 2 - PRODUCTS

2.1 EQUIPMENT LABELS

Identify the selected panelboard, main disconnect, neutral bar, and grounding bar.
Use the package index from Section 01 33 00, Paragraph 2.1 for source references.

2.2 CIRCUIT SCHEDULES

Provide a printed circuit schedule and a matching electronic record.
Identify spare branch positions without assigning a future load.

2.3 FIELD MARKING

Mark conductors at both ends and update the record after field changes.

SECTION 01 78 23 - OPERATION AND MAINTENANCE DATA

PART 3 - EXECUTION

3.1 RECORD COMPILATION

Compile the accepted product package, field test record, labels, and training record.
Retain the final index with the fictional owner representative.

3.2 FINAL REVIEW

Confirm that the package remains searchable and that superseded pages are marked.
Do not treat this closeout review as a substitute for Section 26 24 16 requirements.

3.3 END OF SECTION

This final boilerplate page closes the document and contains no additional cut-sheet requirement.